

Manifolds

Prerequisites

Definition (Lie group). (G, e, m, i) st G smooth, m smooth.

Proposition. The differentials of the maps

- $dm(e, e) : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}, \quad (X, Y) \mapsto X + Y$
- $di(e) : \mathfrak{g} \rightarrow \mathfrak{g}, \quad X \mapsto -X$

Definition (Adjoint).

$$\begin{array}{lll} \mathbf{Ad}(g) : G \rightarrow G & \mathbf{Ad}(g) : \mathfrak{g} \rightarrow \mathfrak{g} & \mathbf{ad} : \mathfrak{g} \rightarrow \text{End}(\mathfrak{g}) \\ h \mapsto ghg^{-1} & X \mapsto gXg^{-1} & X \mapsto [X, \cdot] \end{array}$$

Proposition. $\varphi : G \rightarrow H$. Then

- $d\varphi(e) \circ \mathbf{Ad}(g) = \mathbf{Ad}(\varphi(g)) \circ d\varphi(e)$
- $d\varphi(e)([X, Y]_G) = [d\varphi(e)X, d\varphi(e)Y]_H$

Definition (Lie algebra). Finite dimensional real (or complex) vector space L and bilinear map st

- $[X, Y] = -[Y, X]$
- $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$

Homomorphisms $\varphi : L \rightarrow M$ satisfy $\varphi([X, Y]_L) = [\varphi(X), \varphi(Y)]_M$

Proposition. G top. group, $H \subseteq G$ open subgroup $\implies H$ closed.

Proposition. G connected top. group, $U \subseteq G$ nbh of e Then $G = \bigcup_{n \geq 1} U^n$.

Proposition. Let G top. group, G^0 conn. comp. of 1.

- G^0 closed sbg characteristic (stable under all automorphisms)
- G locally conn. $\implies G^0$ open and contained in every open subgroup
- Conn. comp. of G are cosets G/G^0 .

Definition (Vector field). Assigns $v(p) \in T_p(M)$. Smooth if system of local coord. are smooth functions.

Definition (Integral curve). (I, γ) st $\gamma : I \rightarrow M, \gamma'(t) = c(\gamma(t))$

Definition. $L_g : G \rightarrow G, \quad h \mapsto gh$. Taking differentials $l_g := dL_g(1) : \mathfrak{g} \rightarrow T_g G$

Definition (Left invariant). $v(g) = l_g(v(1))$

Proposition. v left inv. Then, maximal integral curve γ of v with $\gamma(0) = e$ is defined on $\mathbb{R}$ and $\gamma(t+s) = \gamma(t)\gamma(s)$.

Definition. $\exp_G(X) = \gamma_X(1)$, with $\gamma'_X(0) = X$. If V finite dimensional, then

$$\exp_G(X) = \sum_{n=0}^{\infty} \frac{X^n}{n!}$$

Proposition. Properties of $\exp$:

- $\mathbf{Ad}(x) \circ \exp_G = \exp_G \circ \mathbf{ad}(x)$
- $\mathbf{Ad} \circ \exp_G = \exp_{GL(\mathfrak{g})} \circ \mathbf{ad}$
- $d\exp_G(0) = id_{\mathfrak{g}}$
- $\varphi \circ \exp_G = \exp_H \circ d\varphi(1)$
- $\gamma_X(t) = \exp_G(tX)$

Proposition. We have

- $\exp_G : \mathfrak{g} \rightarrow G$ is a local diffeomorphism at 0.
- Image of $\exp_G$ lies in G^0
- $U \subseteq \mathfrak{g}$ nbh of 0, then $\exp(U)$ generates G^0 .
- If G is connected, $\exp_G$ is surjective.
- $f : G \rightarrow H$, then $df_1(X) = \frac{d}{dt} \big|_{t=0} f(\exp_G(tX))$

1 Compact topological groups

Definition (Separation axioms). *These are*

- T_0 : $\forall x \neq y, \exists U$ st $x \in U, y \notin U$ or $y \in U, x \notin U$
- T_1 : $\forall x \neq y, \exists U$ st $x \in U, y \notin U$ and $\exists V$ st $y \in V, x \notin V$
- T_2 : $\forall x \neq y, \exists U, V$ st $x \in U, y \in V, U \cap V = \emptyset$
- T_3 : $\forall x \in X, \forall F \subseteq X$ closed, $x \notin F, \exists U$ open st $x \in U, F \cap U = \emptyset$
- $T_{3\frac{1}{2}}$: $\forall x \in X, \forall F \subseteq X$ closed, $x \notin F, \exists f : X \rightarrow [0, 1]$ st $f(x) = 0, f(F) = \{1\}$
- T_4 : $\forall F, F' \subseteq X$ closed, $F \cap F' = \emptyset, \exists U, V$ st $F \subseteq U, F' \subseteq V, U \cap V = \emptyset$

Proposition. For top. groups, $T_0 \implies T_{3\frac{1}{2}}$.

Proposition (Urysohn's lemma). T_4 is equivalent to $\forall F, F'$ closed $\exists f : X \rightarrow [0, 1]$ continuous, $f(F) = \{0\}, f(F') = \{1\}$.

Proposition. T_2 and compact $\implies T_4$.

Let S a set, Λ a comm. ring

Definition (Algebraic S -Mod). • M S -module over Λ if M is Λ -module and $a = a_M : S \rightarrow \text{End}_\Lambda(M)$

- $N \subseteq M$ S -submodule if N is Λ -submodule and $a(S)(N) \subseteq N$
- $f : M \rightarrow N$ S -morphism if f is Λ -linear and $f(a_M(s)m) = a_N(s)f(m)$ for all $s \in S$

Definition. Algebraic definitions:

- **Irreducible** M has no non-zero proper S -submodules.
- **Indecomposable** M is not a direct sum of two non-zero S -modules.
- **Semisimple** M is a direct sum of irreducible S -modules.

Definition (Length). Supremum of $M_1 \subsetneq M_2 \subsetneq \dots \subsetneq M_n = M$ of S -submodules of M .

Proposition. M semisimple $\iff M$ is a sum of irreducible S -modules

If that happens, then there exists a (non unique) complement for each submodule. If M is finite length, all are equivalent.

Proposition. Let $f : M \rightarrow N$. If M irreducible, then $f = 0$ or f is injective. If N irreducible, then $f = 0$ or f is surjective.

Proposition (Schur's lemma). V, W irreducible fin. dim. $\mathbb{C}$ -vector spaces. Then

$$\dim \text{Hom}_S(V, W) = \begin{cases} 1 & V \cong W \\ 0 & \text{otherwise} \end{cases}$$

Definition. $\hat{S}$ set of isomorphism classes of irreducible S -modules.

Proposition.

$$\bigoplus_{W \in \hat{S}} \text{Hom}_S(W, V) \otimes W \rightarrow V, \quad f \otimes w \mapsto f(w)$$

is an S -morphism. If iso, then V semi-simple. If V semi-simple and all irreducible submod of V are finite, then iso.

Definition. Topological definitions:

- **Irreducible** M has no non-zero proper closed S -submodules.
- **Indecomposable** M is not a direct sum of two non-zero closed S -modules.
- **Semisimple** M is the closure of a direct sum of irreducible S -modules.

Proposition. V finite-dimensional $\mathbb{R}, \mathbb{C}$ -vector space. Let S topological space and $a : S \rightarrow \text{End}(V)$. Then

- a continuous $\iff S \times V \rightarrow V$ continuous
- algebraic and topological definitions agree

Definition. • Repr. of top. group G is a topological G -module (π, V_π) with V_π a $(\mathbb{R}, \mathbb{C})$ -vs, st $\pi : G \rightarrow \text{GL}(V_\pi)$ is a group hom.

- Repr. of Lie algebra L over a field k is a L -module (ρ, V_ρ) with V_ρ a l -vs for some extension l/k , st $\rho : L \rightarrow \mathfrak{gl}(V)$ is a Lie algebra hom.

Definition (Hilbert-Schmidt inner product). $A, B \in \text{End}_{\mathbb{C}}(V)$

$$\langle A, B \rangle_{HS} = \text{tr}(A^*B), \quad \|A\|_{HS} = \sqrt{\langle A, A \rangle_{HS}}$$

Proposition. The isomorphism

$$V \otimes V^* \rightarrow \text{End}(V), \quad v \otimes f \mapsto T_{v,f} := (w \mapsto f(w)v)$$

is an isometry.

We assume $F = \mathbb{R}, \mathbb{C}$

Definition (Top. vs). V vs with topology st $+$ and $\cdot$ continuous.

Definition. V is called

- **Banach** if V complete and comes from a norm $\| - \| : V \rightarrow F$.
- **Hilbert** if Banach and comes from an inner product $\langle -, - \rangle$.
- **Fréchet** if V complete and comes from a countable family of seminorms $\{p_n\}_{n \geq 1}$.
- **Locally Convex** if every nbh of 0 contains a convex nbh of 0. ($\iff$ family of seminorms)
- **Quasi-Complete** if every closed bounded subset is complete.

Theorem (Hahn-Banach). V Hausdorff locally convex top. vs. If $W \subseteq V$ subspace and $\lambda : W \rightarrow F$, there exist an extension on V .

Theorem (Closed Graph). V, W Frechet. $f : V \rightarrow W$ linear is continuous $\iff$ graph of f is closed in $V \times W$.

Theorem (Open mapping theorem). V, W Frechet. $f : V \rightarrow W$ continuous linear surjective $\implies f$ is open.

Proposition. Hilbert $\implies$ Banach $\implies$ Fréchet $\implies$ Locally convex

Definition. $A : V \rightarrow W$ linear btw Banach. A is bounded ($\iff$ continuous) if

$$\|A\| = \sup_{\|v\| \leq 1} \|Av\| < \infty$$

$\text{Hom}_{cts}(V, W)$ is Banach with this norm.

Theorem. X compact top. space with measure μ , V quasi-complete top. vs. Then $f : X \rightarrow V$ continuous $\implies \exists ! v \in V$ st for all $\lambda \in \text{Hom}_{cts}(V, F)$

$$\lambda(v) = \int_X \lambda(f(x)) d\mu(x)$$

Definition (Resolvent set). V Banach, $A \in \text{End}(V)$. Resolvent set $\rho(A) = \{\lambda \in F : A - \lambda \text{id}_V \text{ bijective}\}$, spectrum $\sigma(A) = F \setminus \rho(A)$.

Proposition. $\sigma(A)$ is compact.

Definition. A called compact if the closure of $A(B)$ is compact where B is the unit ball of V .

Definition. For $A \in \text{End}(V)$, there exists A^* st $\langle Av, w \rangle = \langle v, A^*w \rangle$. A self-adjoint if $A = A^*$ and normal if $AA^* = A^*A$.

Theorem (Spectral theorem). A normal compact operator on Hilbert space V . Then

- $\sigma(A)$ is discrete, 0 is the only possible accumulation point.
- $\lambda \in \sigma(A) \setminus \{0\} \implies \lambda$ is an eigenvalue and V_λ fin. dim.
- For $\lambda \neq \mu$, then $V_\mu \perp V_\lambda$
- $V = \hat{\bigoplus}_{\lambda \in \mathbb{C}} V_\lambda$

Theorem (Stone-Weierstrass). X compact Hausdorff, $A \subseteq C(X, \mathbb{C})$ subalgebra st

- $1 \in A$
- A separates points of X ($\forall x \neq y, \exists f \in A$ st $f(x) \neq f(y)$)
- A is closed under complex conjugation

Then A is dense in $C(X, \mathbb{C})$ with respect to the sup. norm.

Theorem (Arzela-Ascoli). X compact Hausdorff, $F \subseteq C(X, \mathbb{C})$. Then the closure of F is compact $\iff$

- F is equicontinuous
- F is pointwise bounded

Theorem. G top. comp. group. There exist a functional

$$\mathcal{C}(G, \mathbb{C}) \rightarrow \mathbb{C}, \quad f \mapsto \int_G f(x) dx$$

- If $f(x) \in \mathbb{R}_{>0}$, then $\int_G f(x) dx > 0$
- $\int_G f(x) dx = \int_G f(gx) dx = \int_G f(xg) dx$ for all $g \in G$
- $\int_G 1 dx = 1$

Definition. V representation of G . Then π is called

- **Banach representation** if V_π is Banach and π preserves the norm.
- **Hilbert representation** if V_π is Hilbert and π preserves the inner product.
- **lcc representation** if V_π is Hausdorff, loc. conv. and complete.

Proposition. G compact, then $\mathcal{G} \subset L^2(G) \subset L^1(G)$

Compact Lie groups

Proposition. G Lie, $\pi : G \rightarrow GL(V)$ a f.d. rep of G . Then

- $d\pi(1) : \mathfrak{g} \rightarrow gl(V)$ is a Lie algebra rep of $\mathfrak{g}$.
- $\pi \rightarrow d\pi(1)$ faithful additive exact functor
- G connected, fully faithful functor, + simply connected, equivalence of categories.

Proposition. f.d. representation π . TFAE

$$W \text{ is } G\text{-stable} \iff W \text{ is } \mathfrak{g}\text{-stable} \iff W_{\mathbb{C}} \text{ is } G\text{-stable} \iff W_{\mathbb{C}} \text{ is } \mathfrak{g}\text{-stable} \iff W_{\mathbb{C}} \text{ is } \mathfrak{g}_{\mathbb{C}}\text{-stable}$$

Definition (Adjoint representation). $\text{ad} : L \rightarrow \mathfrak{gl}(L)$, $x \mapsto [x, \cdot]$. Center $Z(L) = \ker \text{ad}$

Definition (Ideal). $I \subseteq L$ such that $[I, L] \subseteq I$

Example. $\mathfrak{b}, \mathfrak{u}, \mathfrak{t} \subseteq \mathfrak{gl}_n(k)$ upper, strictly upper and diagonal. All subalgebras, no ideals. $\mathfrak{u}$ ideal of $\mathfrak{b}$.

Proposition. $Z(L), [L, L], \ker \rho$ ideals

Definition. L is called

- **Simple** if not abelian and no non-trivial ideals.
- **Semisimple** if direct sum of simple ideals. Also $L = [L, L] = \oplus L_i$ with L_i simple ideals.
- **Reductive** if direct sum of semisimple and abelian ideals. $L = Z(L) \oplus [L, L]$ with $[L, L]$ semisimple.

Definition (Invariant form). $b : V \times V \rightarrow k$ if $b(\pi(x)y, z) = -b(y, \pi(x)z)$. Equivalently,

$$b([x, y], z) = b(x, [y, z])$$

Definition (Killing form). invariant bilinear symmetric $\kappa : L \times L \rightarrow k$, $\kappa(x, y) = \text{tr}(\text{ad } x \circ \text{ad } y)$

Proposition. Killing form of L is definite, then L is semisimple.

Definition (Compact lie algebra). killing form of $L/Z(L)$ is negative definite. Also

- $L/Z(L)$ semisimple
- $Z(L) = \text{rad}(k)$

Proposition. L compact and semi-simple. Then

- $\text{ad} : L \rightarrow \text{Der}(L)$ is isomorphism
- $\text{Ad}(L) = \text{Aut}(L)^\circ$
- $\text{Ad}(L)$ has no center.

Proposition. TFAE

- L is compact
- $\text{Ad}(L)$ is compact
- The image of $t \mapsto \exp(t \cdot \text{ad}(x))$ is bounded for any $x \in L$.
- $\text{ad}(x)$ is diagonalizable with purely imaginary eigenvalues for all $x \in L$.

Proposition. G Lie group with Lie algebra $\mathfrak{g}$.

- G compact $\implies \mathfrak{g}$ compact
- $\mathfrak{g}$ compact and semi-simple, then G compact.

Proposition. L compact $\implies L$ reductive.